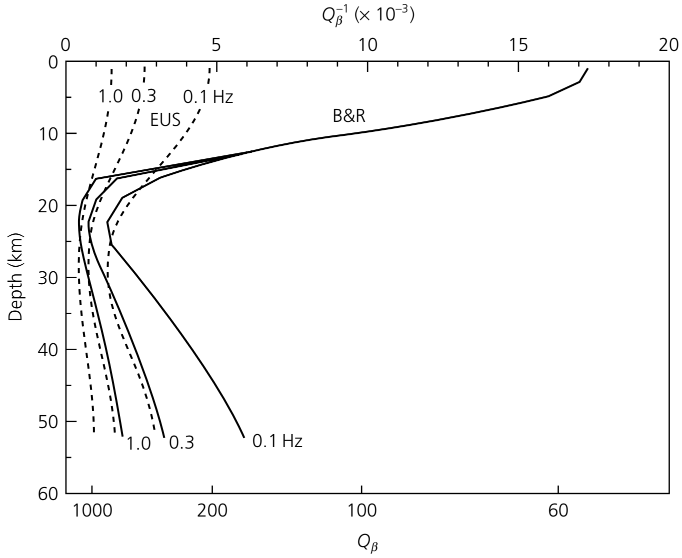


**Fig. 3.7-17** Variation in attenuation with lithospheric depth for the eastern USA (EUS) and Basin and Range (B&R). Lower attenuation occurs for higher frequencies. (Mitchell, 1995. *Rev. Geophys.*, *33*, 441–62, copyright by the American Geophysical Union.)

Regional variations in crustal $Q$ are often studied with $Lg$ waves, a superposition of higher-mode surface waves that give prominent arrivals in continental regions. $Q_{Lg}$ for the USA varies regionally (Fig. 3.7-18), with values as high as 750 in the stable East and as low as 250 in the tectonically active West. This regional difference in attenuation, also seen in Figs 3.7-1 and 3.7-17, has implications for seismic hazards (Section 1.2.2). Similarly, the fact that the USA tested nuclear weapons in the western USA, which is more attenuative than the areas used by the Soviet Union, is significant for verifying test ban treaties (Section 1.2.8).

Attenuation in the upper mantle varies with depth, with the lowest $Q$ in the asthenosphere from about 80 to 220 km depth (Fig. 3.7-19). At these depths the temperature approaches, and perhaps exceeds, the melting temperatures of rock, so a small percentage of partial melt may exist. This pattern of attenuation is similar to that for seismic velocities, which are lowest in the asthenosphere. Hence both the elastic velocity and anelastic attenuation reflect the physical processes causing the mechanically weak asthenosphere. Beneath the asthenosphere, $Q$ increases gradually with depth, presumably because temperature increases at a slower rate than pressure.

$Q_\mu$ increases with depth through the lower mantle, reaching values in excess of 500. There is some indication that attenuation is enhanced in the D″ region at the base of the mantle. Although no attenuation of $P$ waves is detected for the outer core, there is significant attenuation of $PKIKP$ waves traversing the inner core, yielding $Q_K$ estimates in the range of 150–300.

Lateral variations in attenuation are studied using tomographic methods similar to those used for velocity (Sections 2.8.3, 7.3). Where temperatures vary over short distances, significant attenuation variations can occur, as shown in Fig. 3.7-3 for a mid-ocean ridge. Similarly, a cross-section through the back-arc spreading center above the Tonga subduction zone (Fig. 3.7-20) shows that $Q_\alpha$ exceeds 10,000 within the cold and rigid subducting slab, but is less than 75 beneath the hot back-arc basin. Such attenuation data, especially when combined with velocity data, are valuable for tectonic studies.

## 3.8 Composition of the mantle and the core

Seismology yields information about velocities within the earth. To derive inferences about the composition of the earth, the seismological data are combined with results from geology, geodesy, geomagnetism, cosmochemistry, and the physics and

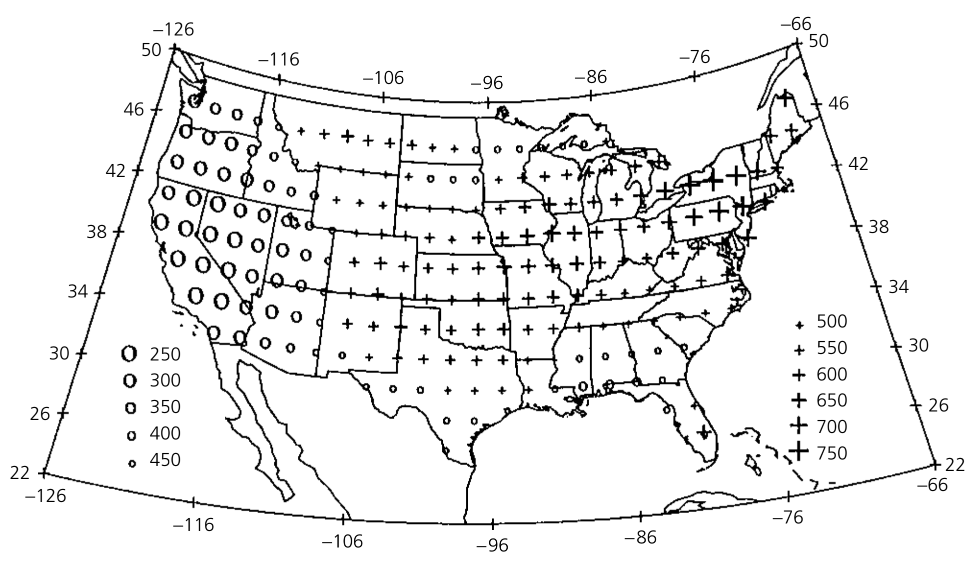


**Fig. 3.7-18** $Q_{Lg}$ for the USA mapped from the codas of 1 Hz $Lg$ waves. $Q_{Lg}$, which reflects attenuation within the crust, shows higher attenuation in the tectonically active western USA and lower attenuation in the tectonically inactive east. (Mitchell *et al.*, 1997.)

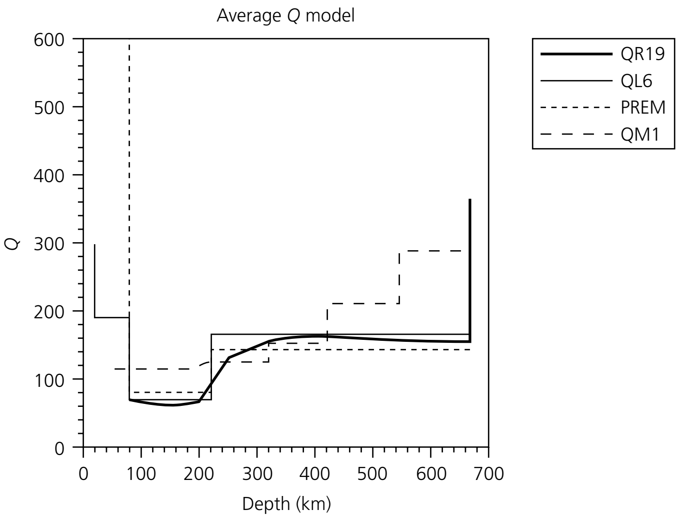


**Fig. 3.7-19** Models of $Q$ in the upper mantle showing that attenuation is highest at 80–220 km depth and then decreases with depth. (Romanowicz, 1995. *J. Geophys. Res.*, *100*, 12,375–94, copyright by the American Geophysical Union.)

chemistry of materials at high temperature and pressure. A general view of the earth's composition has emerged, although aspects are still under investigation. This view is a cornerstone of our thinking about the evolution of the earth and other planets. We will summarize some basic ideas that are presently under discussion, and the suggested readings provide more information.

### 3.8.1 *Density within the earth*

A starting point for analysis of the earth's composition is a model of the variation in density with depth. The density is an important constraint on the nature of the material, and can be combined with velocities to derive elastic constants. Densities are less well known than velocities, and their estimation requires more inferences. As with velocities, we use a radially symmetric density model for most applications and consider lateral perturbations when needed.

The basic constraint on the earth's density is that its average is given by the earth's mass $M$, which can be found from the acceleration of gravity at the surface $r = a$ using the law of gravitation,

$$g = GM/a^2. \tag{1}$$

Because $g = 9.8$ m/s$^2$, $G = 6.67 \times 10^{-11}$ Nm$^2$kg$^{-2}$, and $a = 6371$ km, we find $M = 5.97 \times 10^{24}$ kg. The mass is the volume integral of the density, so if density varies only with depth

$$M = 4\pi \int_0^a \rho(r) r^2 dr, \tag{2}$$

the average density, $\rho_o$, is found by dividing the mass by the volume,

$$\rho_o = M/[(4/3)\pi a^3]. \tag{3}$$

The resulting average density of the earth is about 5.5 g/cm$^3$. The fact that this value is significantly higher than the density of the surface rocks (about 3 g/cm$^3$) is evidence for a core of much denser, and hence presumably different, material.

A second constraint on the density, which also indicates a dense core, comes from the moment of inertia about the rotation axis. This is defined by (Fig. 3.8-1) integrating over volumes $dV$, each at a distance $l = r \sin\theta$ from the spin axis,

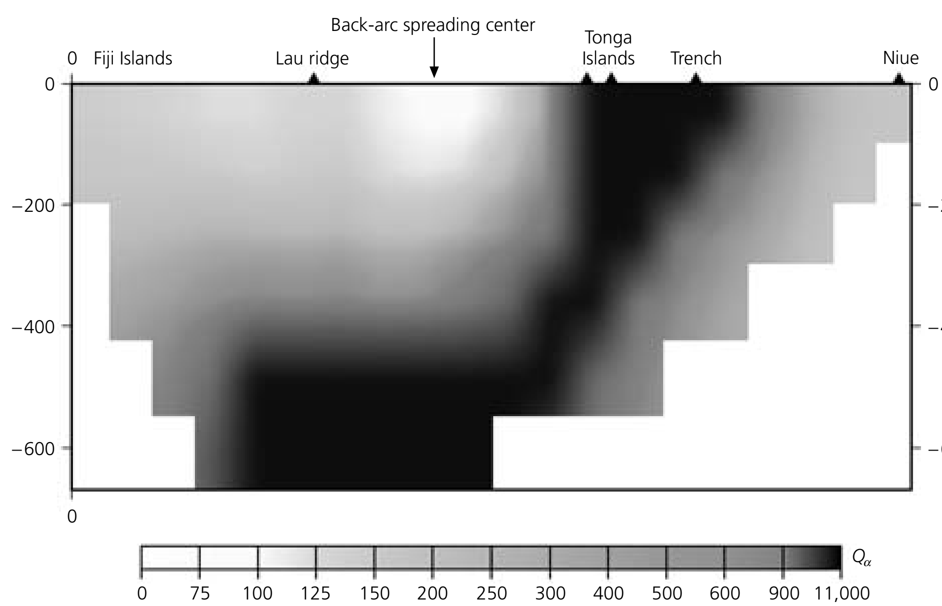


**Fig. 3.7-20** Cross-section across the Tonga subduction zone, showing large lateral variations in $Q_\alpha$ between the cold subducting slab (black) and the hotter back-arc basin. (Roth *et al.*, 1999. *J. Geophys. Res.*, *104*, 4795–809, copyright by the American Geophysical Union.)

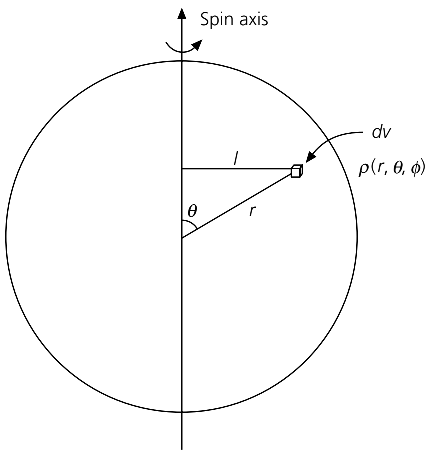


**Fig. 3.8-1** A planet's moment of inertia is found by integrating about the spin axis. The moment arm, $l$, to a volume element, $dV$, is $r \sin \theta$.

$$C = \iiint l^2 \rho(r, \theta, \phi) dV = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \rho(r)(r^2 \sin^2 \theta) r^2 \sin \theta \, dr d\theta d\phi$$

$$= \frac{8}{3}\pi \int_0^a \rho(r) r^4 dr. \tag{4}$$

The ratio of the moment of inertia to the mass gives a scalar that depends on the density distribution. If the earth were homogeneous, the density everywhere would equal the average density, $\rho(r) = \rho_o$, and

$$C = (8/15)\pi a^5 \rho_o, \quad M = (4/3)\pi a^3 \rho_o, \quad C/Ma^2 = 0.4. \tag{5}$$

Alternatively, if all the mass were in a shell at the surface, the density distribution could be written as a delta function $\rho(r) = \delta(r - a)\rho_s$. Using the properties of the delta function (Section 6.2.5), Eqns 2 and 4 yield

$$C = (8/3)\pi \rho_s a^4, \quad M = 4\pi \rho_s a^2, \quad C/Ma^2 = 0.67. \tag{6}$$

As expected, a distribution with material concentrated toward the outside gives a larger ratio.

A more realistic case is a two-shell planet, with a mantle of density $\rho_m$ and a core of density $\rho_c$ and radius $r_c$. The integrals are evaluated in pieces as

$$C = \frac{8}{3}\pi \left[ \int_0^{r_c} \rho_c r^4 dr + \int_{r_c}^{a} \rho_m r^4 dr \right] = \frac{8}{15}\pi [\rho_m a^5 + (\rho_c - \rho_m) r_c^5],$$

$$M = \frac{4\pi}{3} [\rho_m a^3 + (\rho_c - \rho_m) r_c^3]. \tag{7}$$

Values similar to those for the earth ($\rho_c$ = 12 g/cm$^3$, $\rho_m$ = 5 g/cm$^3$, $r_c$ = 3480 km) yield a moment of inertia ratio of $C/Ma^2 = 0.35$. This value is less than the 0.4 which a uniform planet would have, because the material is concentrated toward the center. It is similar to the value of $C/Ma^2$ for the earth determined from the earth's shape and gravity field. The earth's value, about 0.33, thus indicates the presence of a dense core.

Although the mass and moment of inertia give only integral constraints on the density, seismic velocities give information on the variation of density with depth. We first consider a region of uniform material and see how the density increases with depth as the material is *self-compressed* by its own weight. At a radius $r$, the gradient of the hydrostatic pressure $P(r)$ is

$$\frac{dP}{dr} = -g\rho, \tag{8}$$

where $\rho(r)$ and $g(r)$ are the density and the acceleration of gravity at that depth. The derivative is negative, because pressure increases with depth. The local value of gravity, $g(r)$, depends on the total mass $m(r)$ within the sphere of radius $r$,[1]

$$g = Gm/r^2. \tag{9}$$

The pressure derivative can then be written as

$$\frac{dP}{dr} = \frac{-\rho G m}{r^2}. \tag{10}$$

The elastic constants of the material are introduced using the definitions of the density and the dilatation $\theta$ (Eqn 2.3.60),

$$\rho = m/V, \quad d\theta = dV/V, \tag{11}$$

so that differentiation yields

$$d\rho = -(m/V^2)dV = -\rho d\theta. \tag{12}$$

Thus the bulk modulus $K$ can be expressed, starting with its definition (Eqn 2.3.74), as

$$K = -\frac{dP}{d\theta} = -\frac{dP}{d\rho}\frac{d\rho}{d\theta} = \rho \frac{dP}{d\rho}. \tag{13}$$

Combining this with the pressure derivative equation (Eqn 10) gives the change in density with depth

[1] $g(r)$ depends only on the mass below radius $r$, because a spherical shell of uniform density has no net gravitational effect inside the sphere. This situation arises because gravity varies as $r^{-2}$, whereas the shell's mass varies as $r^2$, so larger contributions from the closer portions of the shell are canceled by those from the rest. The fact that a sphere's gravitational attraction is the same as if all its mass were at the center arises in the same way. This effect is not a general property of the center of mass and does not apply for bodies of other shapes. However, it applies for the electric field, which also varies as $r^{-2}$, within a uniformly charged sphere. Deriving this result is said to have delayed Newton for years before presenting the theory of gravitation in 1686. (Feynman *et al.*, 1963.)

$$\frac{d\rho}{dr} = \frac{d\rho}{dP}\frac{dP}{dr} = \frac{-\rho^2 Gm}{Kr^2}. \tag{14}$$

To include the observations of seismic velocities, we define the *seismic parameter*, $\Phi$, and *bulk sound speed*, $\Phi^{1/2}$, such that

$$\Phi = \alpha^2 - (4/3)\beta^2 = K/\rho. \tag{15}$$

Thus we can write the *Adams–Williamson equation* relating the velocity structure to the derivative of density with radius,

$$\frac{d\rho}{dr}(r) = \frac{-\rho(r)Gm(r)}{\Phi(r)r^2} = \frac{-\rho(r)g(r)}{\Phi(r)}, \tag{16}$$

where the dependences on radius are explicitly shown. This equation can be used to estimate the density structure by starting with the near-surface density, using the seismic velocities to find its derivative, and computing the density at a deeper point. The resulting density and value of $g(r)$ are then used in the next step.

However, density increases with depth as a result of mineral phase changes as well as of self-compression, so the Adams–Williamson equation is insufficient. This difficulty was identified in 1936 by K. Bullen, who used the Adams–Williamson approach to find the density throughout the mantle. He then computed the moment of inertia of the mantle and subtracted it from the moment of inertia of the earth, to find the moment of inertia of the core. Figure 3.8-2 shows the $C/Ma^2$ value calculated for the core as a function of the assumed density at the top of the mantle, which is the initial density for the Adams–Williamson calculation. For reasonable values of near-surface density, $\approx 3.3$ g/cm$^3$, the core would have $C/Ma^2$ greater than 0.4, implying that density decreases with depth in the core. This seems unlikely, because the solid inner core should be denser than the liquid outer core. Only implausibly high near-surface densities could cure the problem.

This issue was resolved in the 1950s by F. Birch[2] in a classic series of papers showing that at least one of two assumptions underlying the method was inappropriate. One implicit assumption is that the temperature increases with depth along an *adiabatic gradient*, or "adiabat," such that if a piece of material moves vertically, the pressure-induced temperature change leaves the material at the same temperature as its new surroundings (Eqn 5.4.10). However, the temperature gradient in the mantle is thought to exceed the adiabatic gradient, because a superadiabatic gradient is required for the thermal convection expected in the mantle.[3] The superadiabatic gradient can be included by modifying the Adams–Williamson equation (16) to

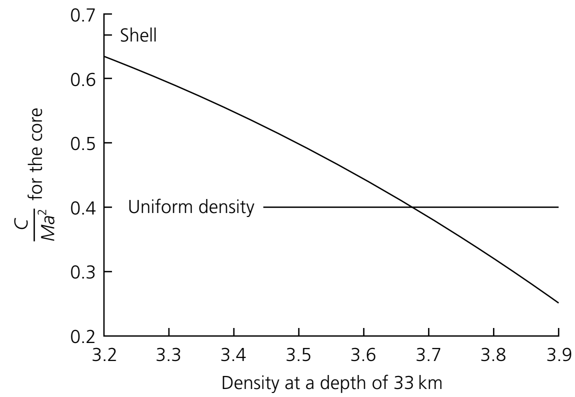


**Fig. 3.8-2** Moment of inertia ratio of the earth's core as a function of density at the top of a uniform mantle. For any realistic upper mantle density the ratio would exceed 0.4, implying that the outer core is denser than the inner core. The alternative is that density increases beyond self-compression occur in the mantle. (After Birch, 1954. *Trans. Am. Geophys. Un.*, *35*, 79–85, copyright by the American Geophysical Union.)

$$\frac{d\rho}{dr} = -\frac{\rho g}{\phi} + g\alpha\tau, \tag{17}$$

where $\alpha$ is the coefficient of thermal expansion,[4] and $\tau$ is the portion of the temperature gradient exceeding the adiabatic gradient. This correction for higher temperature lowers the calculated mantle densities, and hence increases the calculated $C/Ma^2$ for the core, making the problem of the core density structure worse.

Hence the assumption of homogeneous material whose density changes only by self-compression must be incorrect. Birch showed that inhomogeneity can be identified using the function $1 - (1/g)d\phi/dr$. Figure 3.8-3 compares values of this function derived from seismic velocity data with values predicted for compression of homogeneous mantle material. Below 1000 km the mantle behaves as a homogeneous material, while at shallower depths it does not. This is because the mineral phase transitions expected at the 410 and 660 km discontinuities involve denser atomic packings, and therefore transitions to higher densities, than predicted by the Adams–Williamson equation.

As a result, density models of the earth include rapid changes in the transition zone. Figure 3.8-4 shows the velocity and density structure for earth model PREM (Table 3.8-1). Within the lower mantle, outer core, and inner core, density increases smoothly with depth according to the Adams–Williamson equation. At the boundaries between these regions, density

[2] Francis Birch (1903–92) pioneered the use of rock and mineral physics in studies of the earth's composition.

[3] For an adiabatic gradient, rising material reaches the same temperature, and hence density, as its surroundings, and thus has no tendency to continue rising. However, for a superadiabatic gradient, the rising material remains hotter and less dense than its surroundings, and thus tends to continue rising.

[4] The coefficient of thermal expansion, which gives the change in density with temperature $T$, is $\alpha = (-1/\rho)\partial\rho/\partial T$.

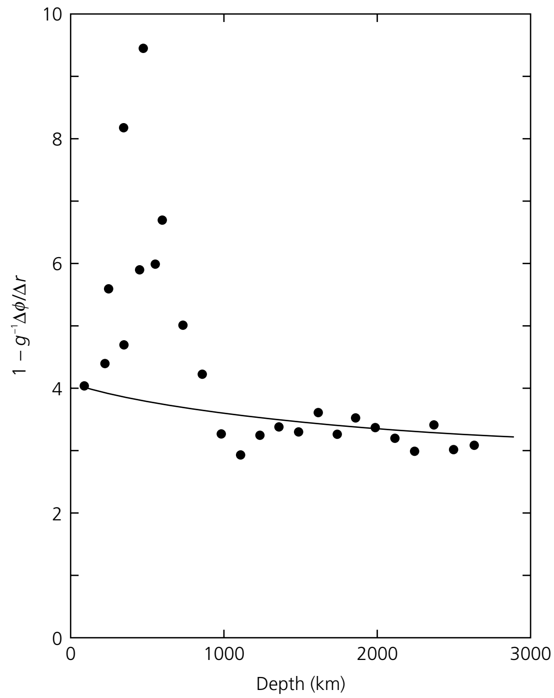


**Fig. 3.8-3** Comparison of observed values (dots) of the function $1 - g^{-1}(\Delta\phi/\Delta r)$ for the mantle, with calculated values (line) of this function for compression of homogeneous material. In the upper mantle transition zone, self-compression alone cannot be occurring, motivating the expectation of mineralogical phase transformations. (After Birch, 1952. *J. Geophys. Res.*, *57*, 227–86, copyright by the American Geophysical Union.)

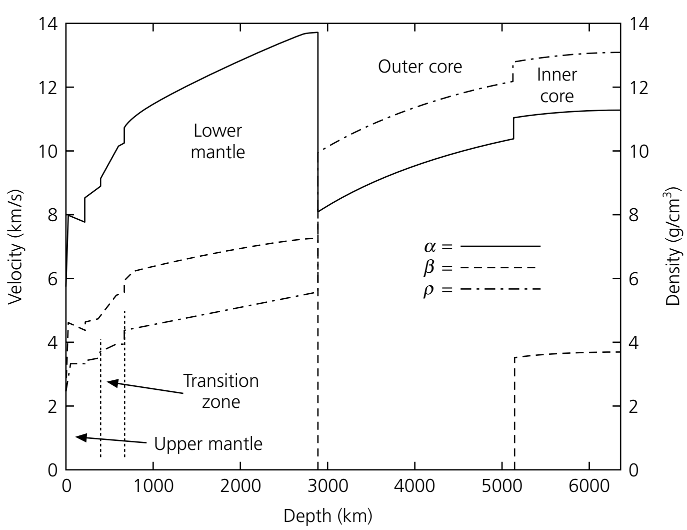


**Fig. 3.8-4** Seismic velocities and density for the Preliminary Reference Earth Model (PREM). (Dziewonski and Anderson, 1981.)

changes sharply. The CMB is the most significant boundary with respect to density, with an increase from 5.57 g/cm$^3$ for mantle rock to 9.90 g/cm$^3$ for the liquid iron outer core. Density also changes sharply at the 410 and 660 km discon-

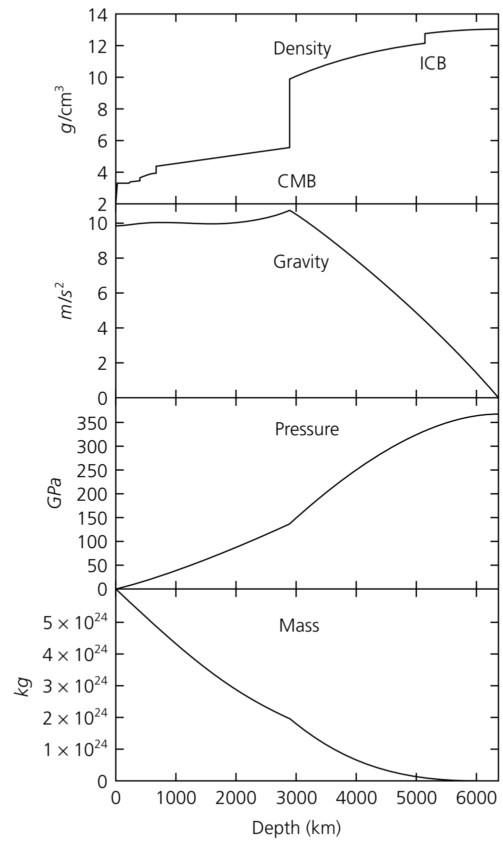


**Fig 3.8-5** Density, gravity, pressure, and mass as functions of depth for the PREM model.

tinuities. Such models are developed to satisfy the travel time data, other seismological data including eigenfrequencies of the earth's normal modes, and the constraints on density.

A density profile lets us compute a pressure profile, and thus use the results of experiments showing which mineral phases exist at particular pressures. To do this we integrate both sides of Eqn 8,

$$P(r) = -\int_0^r g(r)\rho(r)dr, \qquad (18)$$

using $\rho(r)$ and the resulting values of $g(r)$. As shown in Fig. 3.8-5, pressure starts at 1 bar at the surface, and rises to about 13.3 GPa (133 kbar) at the 410 km discontinuity, 23.8 GPa at the 660 km discontinuity, 136 GPa at the CMB, 329 GPa at the ICB, and 364 GPa at the center of the earth.

The curve for gravity is interesting. Gravity averages 9.8 m/s$^2$ at earth's surface,[5] and is zero at earth's center, where the mass of the earth pulls evenly in all directions. Gravity

[5] The value of gravity at the surface is a complicated function, varying laterally as a result of density anomalies within the earth, dynamic forces that lift up or pull down the surface, and a latitudinal effect due to the ellipsoidal shape of the earth (Section A.7.2).

**Table 3.8-1** PREM Model.

| | Depth (km) | $\rho$ (g/cm$^3$) | $\alpha$ (km/s) | $\beta$ (km/s) |
|---|---|---|---|---|
| Ocean | 0.0 | 1.020 | 1.450 | 0.000 |
| | 3.0 | 1.020 | 1.450 | 0.000 |
| Crust | 3.0 | 2.600 | 5.793 | 3.191 |
| | 15.0 | 2.600 | 5.793 | 3.191 |
| | 15.0 | 2.900 | 6.792 | 3.889 |
| | 25.0 | 2.900 | 6.792 | 3.889 |
| Upper mantle | 25.0 | 3.381 | 8.101 | 4.479 |
| | 40.0 | 3.379 | 8.091 | 4.473 |
| | 60.0 | 3.377 | 8.079 | 4.465 |
| | 80.0 | 3.375 | 8.067 | 4.457 |
| | 80.0 | 3.375 | 8.005 | 4.377 |
| low-velocity zone | 115.0 | 3.371 | 7.984 | 4.363 |
| | 150.0 | 3.367 | 7.963 | 4.350 |
| | 185.0 | 3.363 | 7.942 | 4.338 |
| | 220.0 | 3.359 | 7.920 | 4.325 |
| | 220.0 | 3.436 | 8.519 | 4.589 |
| | 265.0 | 3.463 | 8.606 | 4.620 |
| | 310.0 | 3.490 | 8.692 | 4.651 |
| | 370.0 | 3.516 | 8.778 | 4.683 |
| | 400.0 | 3.543 | 8.865 | 4.714 |
| Transition zone | 400.0 | 3.724 | 9.092 | 4.874 |
| | 450.0 | 3.787 | 9.347 | 5.019 |
| | 500.0 | 3.850 | 9.601 | 5.163 |
| | 550.0 | 3.913 | 9.856 | 5.307 |
| | 600.0 | 3.976 | 10.111 | 5.451 |
| | 635.0 | 3.984 | 10.165 | 5.478 |
| | 670.0 | 3.992 | 10.219 | 5.505 |
| Lower mantle | 670.0 | 4.381 | 10.727 | 5.913 |
| | 721.0 | 4.412 | 10.885 | 6.061 |
| | 771.0 | 4.443 | 11.040 | 6.207 |
| | 871.0 | 4.504 | 11.219 | 6.277 |
| | 971.0 | 4.563 | 11.390 | 6.344 |
| | 1071.0 | 4.621 | 11.552 | 6.407 |
| | 1171.0 | 4.678 | 11.707 | 6.469 |
| | 1271.0 | 4.735 | 11.856 | 6.527 |
| | 1371.0 | 4.790 | 11.998 | 6.583 |
| | 1471.0 | 4.844 | 12.135 | 6.637 |
| | 1571.0 | 4.898 | 12.266 | 6.689 |
| | 1671.0 | 4.951 | 12.394 | 6.739 |
| | 1771.0 | 5.003 | 12.518 | 6.788 |
| | 1871.0 | 5.055 | 12.638 | 6.836 |
| | 1971.0 | 5.106 | 12.757 | 6.882 |
| | 2071.0 | 5.157 | 12.873 | 6.928 |
| | 2171.0 | 5.207 | 12.988 | 6.973 |
| | 2271.0 | 5.257 | 13.103 | 7.017 |
| | 2371.0 | 5.307 | 13.218 | 7.061 |
| | 2471.0 | 5.357 | 13.333 | 7.106 |
| | 2571.0 | 5.407 | 13.450 | 7.150 |
| | 2671.0 | 5.457 | 13.568 | 7.195 |
| | 2741.0 | 5.491 | 13.652 | 7.227 |
| | 2771.0 | 5.506 | 13.659 | 7.226 |
| | 2871.0 | 5.556 | 13.684 | 7.226 |
| | 2891.0 | 5.566 | 13.689 | 7.225 |
| Outer core | 2891.0 | 9.903 | 8.065 | 0.000 |
| | 2971.0 | 10.029 | 8.199 | 0.000 |
| | 3071.0 | 10.181 | 8.360 | 0.000 |
| | 3171.0 | 10.327 | 8.513 | 0.000 |
| | 3271.0 | 10.467 | 8.658 | 0.000 |
| | 3371.0 | 10.602 | 8.795 | 0.000 |
| | 3471.0 | 10.730 | 8.926 | 0.000 |
| | 3571.0 | 10.853 | 9.050 | 0.000 |
| | 3671.0 | 10.971 | 9.167 | 0.000 |
| | 3771.0 | 11.083 | 9.278 | 0.000 |
| | 3871.0 | 11.191 | 9.384 | 0.000 |
| | 3971.0 | 11.293 | 9.484 | 0.000 |
| | 4071.0 | 11.390 | 9.579 | 0.000 |
| | 4171.0 | 11.483 | 9.668 | 0.000 |
| | 4271.0 | 11.571 | 9.754 | 0.000 |
| | 4371.0 | 11.655 | 9.835 | 0.000 |
| | 4471.0 | 11.734 | 9.912 | 0.000 |
| | 4571.0 | 11.809 | 9.985 | 0.000 |
| | 4671.0 | 11.880 | 10.055 | 0.000 |
| | 4771.0 | 11.947 | 10.123 | 0.000 |
| | 4871.0 | 12.010 | 10.187 | 0.000 |
| | 4971.0 | 12.069 | 10.249 | 0.000 |
| | 5071.0 | 12.125 | 10.309 | 0.000 |
| | 5149.5 | 12.166 | 10.355 | 0.000 |
| Inner core | 5149.5 | 12.764 | 10.987 | 3.434 |
| | 5171.0 | 12.775 | 10.995 | 3.440 |
| | 5271.0 | 12.825 | 11.030 | 3.465 |
| | 5371.0 | 12.871 | 11.063 | 3.487 |
| | 5471.0 | 12.912 | 11.092 | 3.508 |
| | 5571.0 | 12.949 | 11.119 | 3.526 |
| | 5671.0 | 12.982 | 11.142 | 3.542 |
| | 5771.0 | 13.010 | 11.162 | 3.556 |
| | 5871.0 | 13.034 | 11.179 | 3.568 |
| | 5971.0 | 13.054 | 11.193 | 3.578 |
| | 6071.0 | 13.069 | 11.204 | 3.585 |
| | 6171.0 | 13.080 | 11.212 | 3.590 |
| | 6271.0 | 13.086 | 11.217 | 3.594 |
| | 6366.0 | 13.088 | 11.218 | 3.595 |
| | 6371.0 | 13.088 | 11.218 | 3.595 |

*Source*: Dziewonski and Anderson (1981).

increases slightly across the mantle, reaching a maximum of 10.7 m/s$^2$ at the CMB because of the high density of the core relative to the mantle. Inside the core, gravity decreases nearly linearly toward the earth's center. The high density of the core is also shown by the mass distribution; the core has only 16% of the earth's volume, but has almost one-third of the mass.

### 3.8.2 *Temperature in the earth*

Seismology gives insight into the *geotherm*, the temperature as a function of radius, which both controls and reflects the composition, mineralogy, and evolution of the earth. The geotherm depends on the sources of heat and modes by which the heat is transferred upward in the earth. Thermal convection, heat transfer by the motions of material due to the density changes resulting from temperature, occurs in the mantle. The most obvious manifestations of this convection are the mid-ocean ridges, which are its hot upwelling limbs, and subducting plates, which are its cold downwelling limbs. A separate convection system in the fluid outer core is believed to cause the earth's magnetic field. In addition, heat is transferred by conduction through the lithosphere, the core–mantle boundary, and the inner core, which may also be convecting.

The geotherm is harder to estimate than the pressure profile and remains a subject of debate. A geotherm is inferred by modeling radioactive generation of heat in the crust and the mantle, conduction of heat across the lithosphere, CMB, and inner core, and adiabatic temperature gradients associated with convection in the mantle and the outer core. The predicted temperatures are required to match the expected temperatures of the phase transitions in the transition zone and the expected freezing point of iron at the ICB.[6] Given the uncertainties involved, estimates of the temperature at the center of the earth vary from 5000 K to almost 7000 K,[7] with recent work favoring the lower end of this range.

A sample geotherm for the mantle is shown in Fig. 3.8-6. The most striking feature is the contrast between the shallow temperature gradient in the mid-mantle and the steep gradients in the upper and lower thermal boundary layers, the lithosphere and D″. The difference reflects the assumptions that heat is conducted primarily through the boundary layers, giving the steep gradients, but is convected between them, yielding a shallower near-adiabatic gradient. The predicted temperature rises from about 0°C at the surface to about 1300°C at a depth of 100 km, giving an average thermal gradient of 13°C/km. From there to the base of the mantle the temperature rises only another 1600°C, corresponding to a low gradient of only about 0.6°C/km. Over the bottom few hundred kilometers of the mantle, however, the temperature rises another 1400°C to a CMB temperature of about 4000°C (~3700 K). Thus the temperature changes across the boundary layers at the surface and CMB are comparable. However, because the surface area of the CMB is only about 30% of the earth's surface, much more heat flows out of the earth than flows out of the core. Most of this extra heat is generated by the decay of radioactive isotopes in the mantle and the crust. An important caveat is that if there are additional thermal boundary layers in the mantle, or if the thermal conductivity of the mantle is higher than expected, the temperatures in the lower mantle will be elevated, and the temperature change across D″ will be less.

The geotherm gives insight into the variations with depth of seismic velocity and attenuation and the strength, or stress, the material can support (Section 5.7). Higher temperatures reduce seismic velocity and strength, but increase attenuation. Conversely, higher pressures increase the velocity and strength, but reduce attenuation. These properties thus depend on the balance between the temperature and the pressure. The cold lithosphere has high velocity and low attenuation, and behaves as rigid plates. However, the rapidly increasing temperature with depth brings the geotherm close to, if not above, the *solidus*, or melting temperature curve. This yields the low-velocity zone, where there is high attenuation and weak material that forms the asthenosphere underlying the moving plates. In the lower mantle, temperatures are only slightly greater than in the asthenosphere, so the higher pressures make the rock stronger. Hence the lower mantle is thought to have a viscosity that is about 100 times greater than that in the upper mantle. Temperatures increase rapidly in D″, causing velocities slower than expected from the lower mantle velocity gradient. The ultra-low-velocity zone at the base of the mantle may be due to partial melting, showing that the geotherm has intersected the solidus. As discussed later, the high temperatures in the core keep the outer core liquid, but the rapid increase in pressure due to the weight of the outer core makes the inner core freeze into a denser solid. The inner core is therefore close to the melting temperature of iron, so it has low shear velocities.

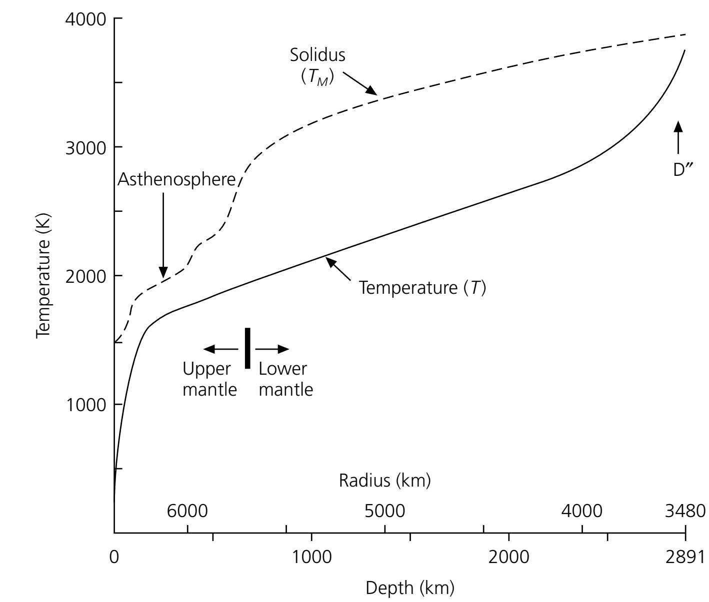


**Fig. 3.8-6** A sample mantle geotherm with steep temperature gradients in the thermal boundary layers at the top and bottom of the mantle and a near-adiabatic gradient in the lower mantle. The melting curve, or solidus, is also shown. Temperatures are given in absolute temperature. (Stacey, 1992. From *Physics of the Earth*, 3rd edn, copyright © 1992 by John Wiley & Sons, Inc. (New York). Reprinted by permission.)

### *3.8.3 Composition of the mantle*

Models of the composition of the mantle are derived by comparing the velocity and density (and therefore pressure) profiles

[6] Although our instincts based on water make it strange to think of temperatures near 5000° as "freezing," this occurs as the solid inner core forms from the liquid outer core.

[7] Temperatures in the deep earth are often given as absolute (Kelvin) temperatures, equal to the Celsius temperatures plus 273.15°.

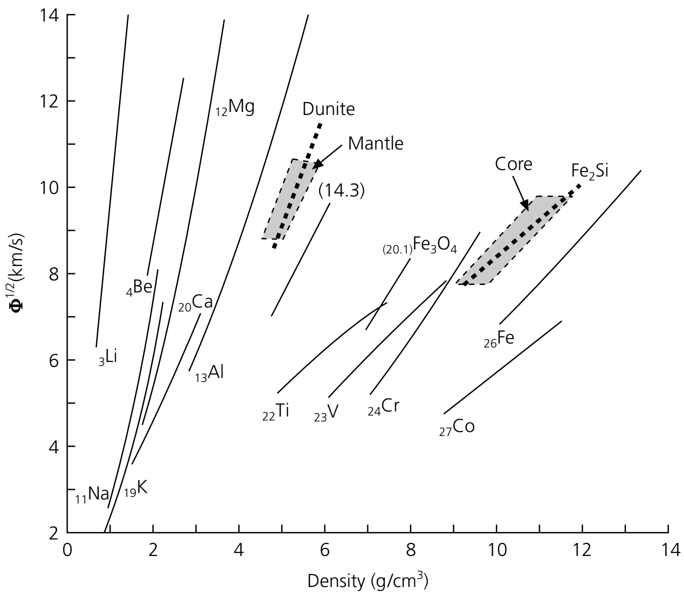


**Fig. 3.8-7** Bulk sound speed as a function of density for various materials, obtained from experiments (lines) compared to the range for the mantle and the core from seismic observations and density models (shaded). Also shown are the results for a dunite rock and the composition $Fe_2Si$. The numbers shown are mean atomic numbers. (After Birch, 1968. *Phys. Earth Planet. Inter.*, *1*, 141–7, with permission from Elsevier Science.)

derived from seismic data to temperature profiles and results for earth materials at high pressure and temperature. A key result from experiments is that the bulk sound speed (Eqn 15) and the density for a material are approximately linearly related for a given mean atomic weight. The mean atomic weight is the mean molecular weight of a formula unit, such that forsterite (magnesian olivine) $Mg_2SiO_4$ has $\bar{m} = (2 \times 24 + 28 + 4 \times 16)/7 = 20$, and fayalite (iron olivine) $Fe_2SiO_4$ has $\bar{m} = (2 \times 56 + 28 + 4 \times 16)/7 = 29$. Figure 3.8-7 shows this result for various elements whose atomic numbers[8] are labeled. Also shown are ranges of density and bulk sound speed for the mantle and core derived from seismically based models. The mantle and the core occupy different parts of the plot.

This result suggests that the mantle and the core are chemically different, and provides a way of testing which chemical compositions are plausible. Dunite, a rock containing 92% olivine, which in turn is 90% forsterite, fits the mantle data. Curves for more iron-rich olivine would plot further to the right, such that olivine with more than 50% fayalite would be outside the range observed for the mantle.

The core data plot much further to the right, indicating that the core is composed of material of higher atomic number. The data are to the left of the curve for pure iron, suggesting that the core is composed of iron plus a lower atomic weight ("lighter") element. For example, the composition $Fe_2Si$ (iron plus 20% weight Si) fits the core data.

[8] The atomic number is the number of protons, whereas the atomic weight is the number of protons and neutrons.

**Table 3.8-2** Pyrolite model, mineralogy above transition zone.

| Mineral | Composition | wt(%) |
|---|---|---|
| Olivine ($Fo_{89}$) | $(Mg_{0.89}, Fe_{0.11})_2SiO_4$ | 57 |
| Orthopyroxene | $(Mg, Fe)SiO_3$ | 17 |
| Clinopyroxene | $(Ca, Mg, Fe)_2Si_2O_6 - NaAlSi_2O_6$ | 12 |
| Pyrope-rich garnet | $(Mg, Fe, Ca)_3(Al, Cr)_2Si_3O_{12}$ | 14 |

*Source*: Ringwood (1979).

Various chemical models for the mantle have been proposed. The concepts involved can be illustrated by considering a proposed composition called *pyrolite* that satisfies various petrological, cosmochemical, and geophysical constraints. Pyrolite is similar to natural peridotites (Fig. 3.2-23), which are acceptable source rocks for basaltic magmas that result from partial melting of mantle rock. The variation in seismic velocity and density with depth is assumed to result from transformations to denser phases as a result of increased pressure. Table 3.8-2 gives a composition whose density at surface temperature and pressure conditions would be 3.38 g/cm$^3$ and has *P*- and *S*-wave velocities consistent with those observed for the upper mantle.

In the upper mantle, the model's major mineral component is olivine. Such a composition satisfies the density and bulk sound speed data (Fig. 3.8-7) and is consistent with the observed seismic anisotropy (Fig. 3.6-4). The transition zone corresponds to a series of solid state phase changes (Fig. 3.8-8). Olivine undergoes several transformations before converting to a *perovskite* structure in the lower mantle. Pyroxene first transforms to garnet, and somewhat deeper, the calcium-bearing component of the garnet transforms to a perovskite structure. Because of the predicted predominance of perovskite (~70%) in the voluminous lower mantle, perovskite is the most abundant material in the earth.

Figure 3.8-9 shows the predicted volume fraction of the major mineral phases as a function of depth. The $\alpha$ phase of olivine, which occurs in the crust and the upper mantle, transforms with increased pressure to its $\beta$ phase *wadsleyite*, which has a modified spinel structure. This transformation is observed experimentally to occur at a pressure of about 12 GPa (120 kbar), corresponding to the 410 km discontinuity. The $\beta$ phase transforms to a $\gamma$, or spinel, structure known as *ringwoodite* (Fig. 3.8-10) at a pressure of ~15 GPa, corresponding to the less dramatic seismic discontinuity at 520 km. At pressures above about 24 GPa, corresponding to the 660 km discontinuity, $\gamma$ spinel breaks down to a perovskite structure and (Mg, Fe)O *magnesiowustite*.

The $(Mg,Fe)SiO_3$ pyroxene component also undergoes changes, beginning with a transformation to garnet below about 200 km. Below 600 km, some of the *Mg*-bearing garnet, majorite, transforms to a structure called *ilmenite*. Beneath about 660 km, the majorite/ilmenite transforms to perovskite. Some of the

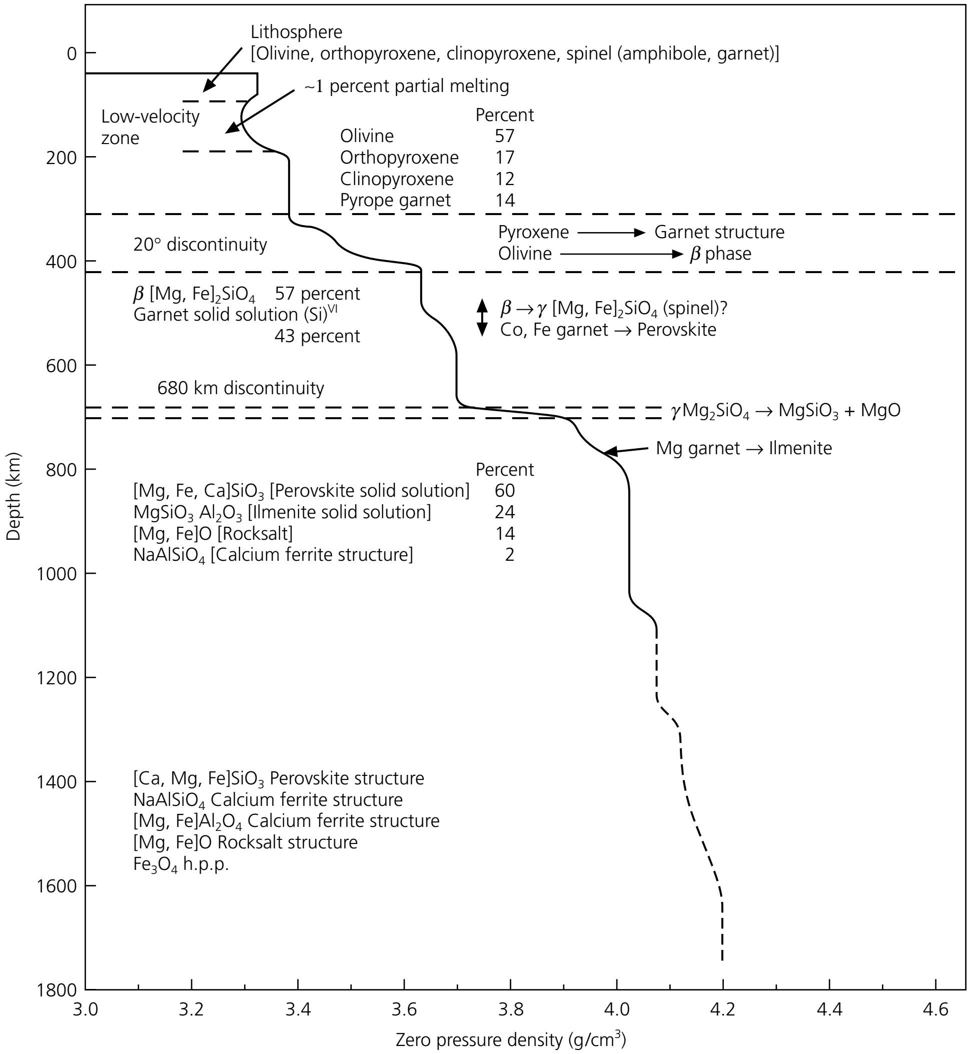


**Fig. 3.8-8** Predicted mineral assemblages as a function of depth for a mantle of pyrolite composition. (Ringwood, 1979. Composition and origin of the earth, in *The Earth, Its Origin, Structure and Evolution*, ed. M. W. McElhinny, copyright 1979 by Academic Press, reproduced by permission of the publisher.)

majorite probably survives into the lower mantle as *stishovite*, a high-pressure phase of quartz, and an $Al_2O_3$-rich phase. Unlike the olivine transformations that cause distinct seismic discontinuites in the transition zone, the pyroxene and garnet transformations occur gradually and contribute to a high velocity gradient through the transition zone down to about 770 km (Section 3.5.4).

These phase changes are investigated using experiments that simulate the pressures, temperatures, and compositions in the earth. Because the experiments are difficult, extrapolations of lower pressure and temperature data via thermodynamic calculations are also used. An important factor for the velocity structure is that some phase transformations happen gradually over a range of depths (Fig. 3.8-11). A simple *univariant* phase change, in which material of a single composition changes completely from one phase to another as pressure increases, causes a sharp discontinuity in velocity. A more complicated *multivariant* phase change involving a system of variable compositions causes two or more phases to coexist over a broad region of pressure, and so produces a velocity gradient. Thus seismological studies that better define the velocity structure of the transition zone improve our understanding of its composition.

The mineralogical models agree with the depths of the seismic discontinuities and their other characteristics. The olivine $\alpha$-to-$\beta$ reaction should occur over a narrow depth range, as shown by the volume fractions in Fig. 3.8-9. This prediction is consistent with the sharpness of the seismic discontinuity, which is observed with high-frequency (short-wavelength) waves. The transformation is exothermic (releasing heat) and hence would occur at lower pressures in subducting slabs due to the colder temperatures (Section 5.4.2). This expectation agrees with seismic observations showing an elevation of the 410 km discontinuity in and around subducting lithosphere. By contrast, the $\beta$-to-$\gamma$ transformation should occur over a broader depth range. This prediction agrees with seismic observations of the 520 km discontinuity, which is invisible to high-frequency waves and seen only with longer wavelengths.

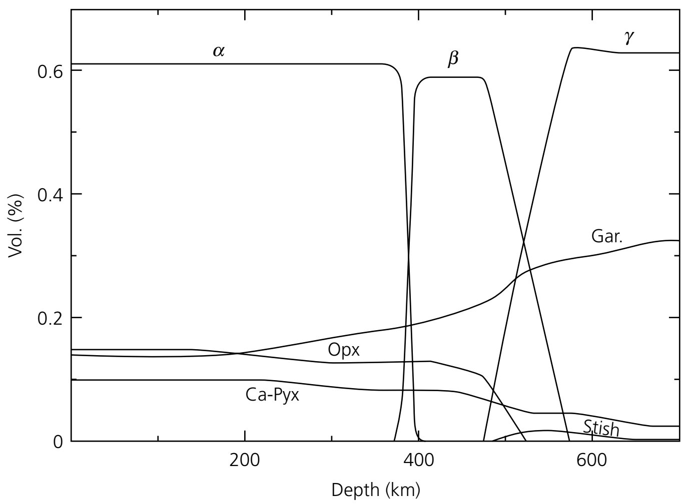


**Fig. 3.8-9** A model for the relative proportions of major mineral phases as a function of depth in the upper mantle. The rapid changes between the olivine and spinel phases ($\alpha$, $\beta$, $\gamma$) cause seismic discontinuities at depths of 410 km and 520 km, whereas the gradual transformation of pyroxene to garnet steepens the velocity gradient in the transition zone (410 km to 660 km). (Weidner, 1986. Reproduced with permission of Springer-Verlag.)

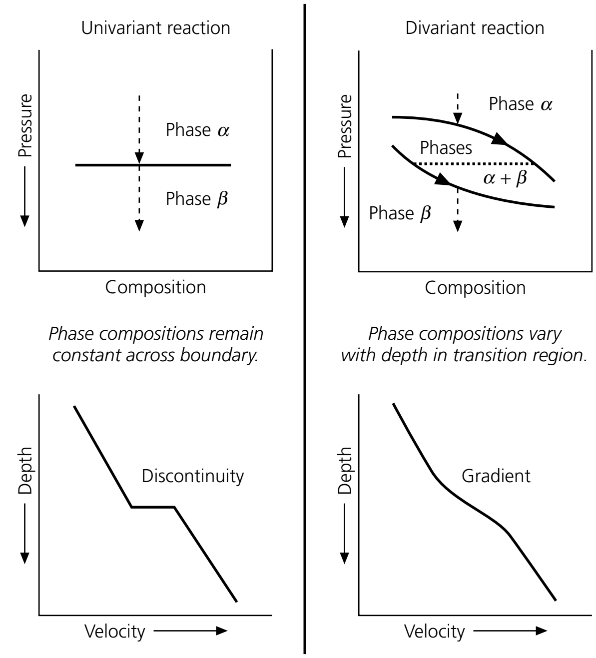


**Fig. 3.8-11** Schematic phase diagrams showing the relation between the nature of a phase change and the corresponding velocity discontinuity. (Bina and Wood, 1987. *J. Geophys. Res.*, *92*, 4853–66, copyright by the American Geophysical Union.)

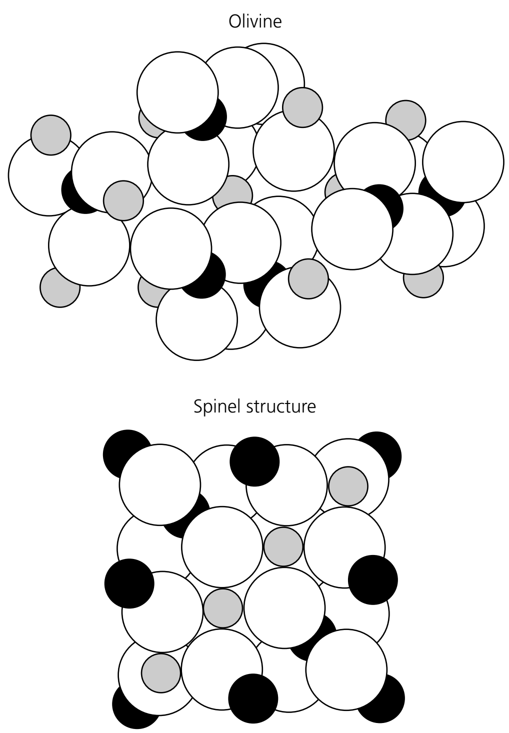


**Fig. 3.8-10** Comparison of the crystal structures of $(Mg, Fe)_2SiO_4$, in its low-pressure $\alpha$ olivine phase (*top*) and its $\gamma$-spinel ringwoodite phase (*bottom*), which is about 10% denser. Spheres correspond to ions of oxygen (white), silicon (black), and magnesium/iron (grey). (After Press and Siever, 1982.)

However, the $\gamma$-spinel to perovskite and magnesiowustite transition should occur over a narrow depth range, consistent with the observed sharpness of the 660 km seismic discontinuity. The reaction is endothermic (absorbs heat) and so should occur at greater depths for colder temperatures. Studies have shown that the discontinuity is depressed to depths of 700 km or more in and around subducting lithosphere.

An unresolved question is whether the lower mantle is chemically distinct from the upper mantle, which has important implications for how the two have mixed during the earth's evolution. In models like those depicted in Fig. 3.8-8, the two are assumed to have the same bulk chemistry, and the increasing velocity and density in the lower mantle result from self-compression. The velocity data do not appear to require phase changes in the lower mantle. However, the lower mantle may be denser than expected for pyrolite, and hence perhaps enriched in iron and silica. The observation that some subducting lithosphere penetrates the 660 km discontinuity (Section 5.4) indicates that mixing occurs. However, even if all slabs reach the lower mantle, the earth may not be old enough for the lower and upper mantles to be well mixed.[9] Another possibility is that the early earth had distinct upper and lower mantle convection systems, and whole mantle convection began later.

[9] Consider a bowl of cake batter after only a few beats of a mixing spoon.

### 3.8.4 Composition of D″

Seismic observations give a picture of the D″ region (Section 3.5.4) that includes lateral velocity variations, vertical layering, and anisotropy. Hence processes there may be as complex as in the lithosphere, the other major thermal boundary layer. This complexity may reflect factors including subducted lithosphere, the generation of mantle plumes, and interactions between the core and the mantle.

Figure 3.8-12a (*right*) shows a simple convection model, with cold material sinking to the CMB, heating up from contact with the core, and then rising again. The *left* side of the figure shows the resulting vertical velocity profiles in regions of downwelling (solid line) and upwelling (dashed line). Thus the large (> ±5%) lateral seismic variations at the base of the mantle would be caused by temperature variations. However, given the complex seismic structures observed, this model component seems necessary but insufficient.

The other possibilities shown involve subducted slabs. In Fig. 3.8-12b, the subducted slabs do not reach the top of the core, but remain separated by a chemically distinct layer. This layer may result from early planetary differentiation, or may have grown by chemical reactions between the mantle and the core. High-pressure experiments imply that perovskite and magnesiowustite would react with iron. These mantle dregs might be thinned in regions of mantle downwelling, and thickened beneath upwellings. Layering in the dregs may explain observations of transverse isotropy in downwelling regions and azimuthal anisotropy in upwelling regions (Section 3.6.6). The velocity increase of the D″ discontinuity may be partly caused by ponded slab material, which will still be colder and have higher velocity than ambient rock. This discontinuity may be enhanced by dregs flowing up and over ponded slabs. The ultra-low-velocity zone (ULVZ) at the very bottom of the mantle may be due to the lower velocities of an iron-rich layer or to partial melting within it.

Another possibility is that the part of the subducted lithosphere that started as basaltic ocean crust and then transformed to eclogite transforms to a material that is seismically faster than the rest of the lower mantle (Fig. 3.8-12c). This phase could delaminate from the slabs and accumulate, forming a different chemical boundary layer. If it remained solid, it might partially explain the D″ discontinuity. Alternatively, if it melted, it might explain the ULVZ. Either way, its laminar nature might explain the observed seismic anisotropy. The lateral variations in velocity would correlate with anisotropy; *SH* waves would travel fast in downwelling regions because of transverse isotropy, but be slowed by the vertical laminations beneath upwellings.

D″ may also signify the bottom of the perovskite stability field (Fig. 3.8-12d). Large radial changes in temperature and/or composition at the base of the mantle could move perovskite or a secondary phase out of its range of stability, causing a phase transformation. One possibility is a transformation of perovskite to stishovite and magnesiowustite, which occurs with an increase in the iron/magnesium ratio. Stishovite has high seismic velocities and might contribute to the D″ discontinuity. In this case, anisotropy might reflect orientation of crystals due to lateral flow. The denser magnesiowustite might settle to the bottom, forming the ULVZ.

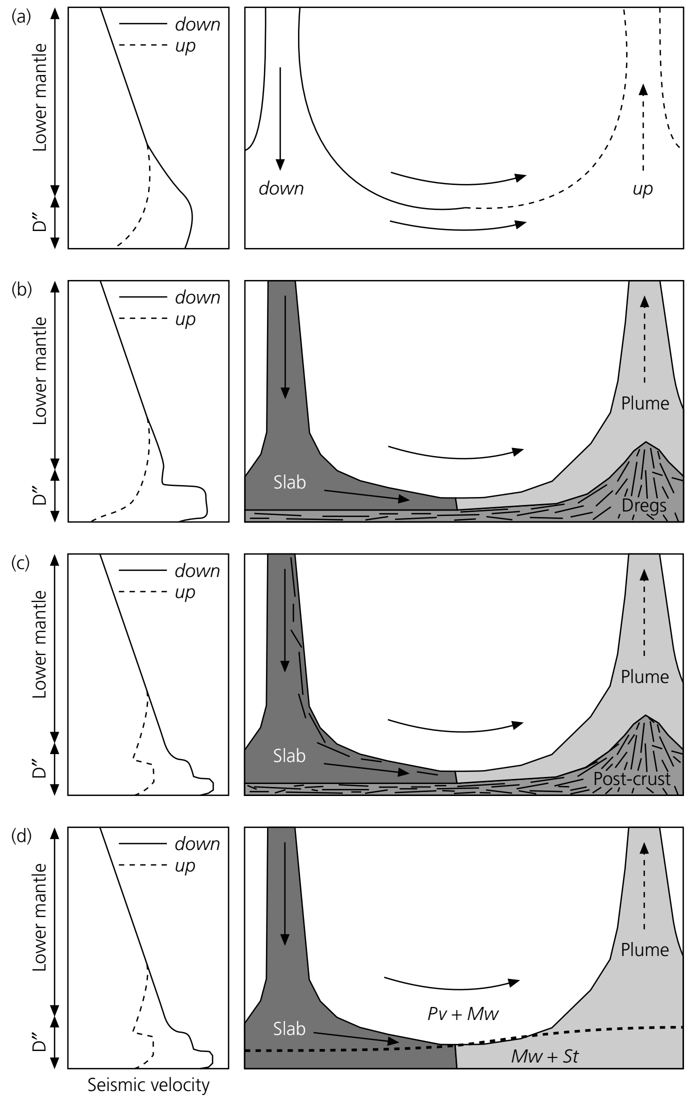


**Fig. 3.8-12** Schematic diagram of processes that might cause the velocity structures observed at the base of the mantle. *Right* panels show the different scenarios, and *left* panels show the resulting velocity–depth profiles for regions of downwelling (solid lines) and upwelling (dashed lines). The scenarios, discussed in the text, are (a) general thermal convection, (b) the interaction of subducted slabs with a chemical boundary layer consisting of dense mantle dregs, (c) a chemical boundary layer formed from delaminated post-eclogitic ocean crust brought down with the slabs, and (d) a mineralogical phase change. (After Wysession, 1996a. *Subduction*, 369–84, copyright by the American Geophysical Union.)

Given our limited knowledge, D″ may involve these and other effects. For example, if the vertical temperature difference across D″ is small (about 300°C), then convection should play

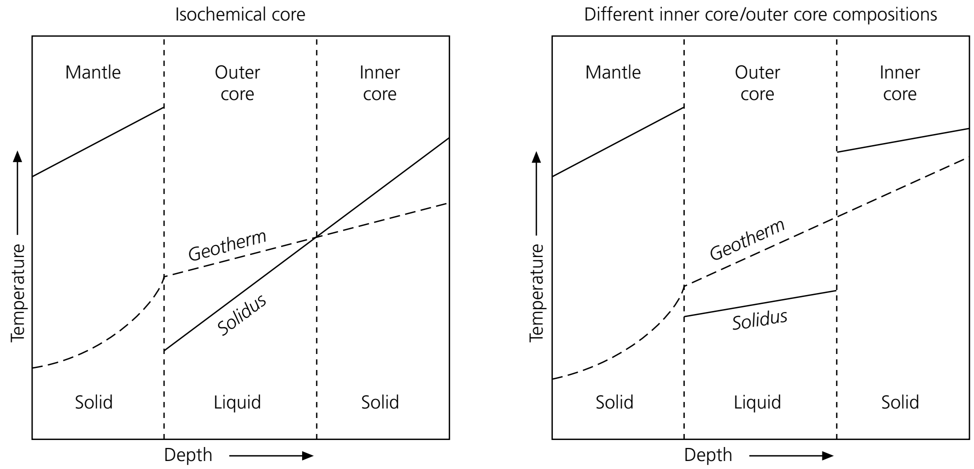


**Fig. 3.8-13** Possible relationships between the geotherm (dashed line) and the solidus (solid line), for the inner and outer cores. *Left*: If the core is homogeneous, the solidus should be continuous across the inner and outer cores, so the gradient of the geotherm must be shallower than that of the solidus for the inner core to be solid and the outer core to be liquid. *Right*: If the inner and cores are chemically different, the solidus can differ between them, allowing a steeper gradient for the geotherm.

a lesser role relative to a chemical boundary layer. If the contrast is large, perhaps 1500°C, plume generation should be more significant, and it would be harder to maintain a distinct chemical layer.

### *3.8.5 Composition of the core*

Interesting issues about the core also remain unresolved. The density and bulk sound speed data (Fig. 3.8-7) suggest that the core has a composition similar to that of iron, but with a less dense element of lower atomic number added. Other arguments for an iron core are from cosmochemistry. Meteorites are roughly divided into stony meteorites, resembling the mantle, and iron meteorites, composed of an iron–nickel alloy, which are thought to be similar to the core.[10] Convection of molten iron is also considered the only suitable mechanism for generating the earth's magnetic field. The light element lowering the core density is unknown: candidates include sulphur, silicon, oxygen, potassium, and hydrogen. Laboratory experiments suggest that 10–15% of a light element would yield an acceptable density.

It may seem surprising that the inner core is solid, because it should be at a higher temperature than the liquid outer core. Thus the effects of pressure favoring the denser solid phase must exceed those of temperature. From the ICB to the center of the earth, temperature is thought to increase by only 100–200°C, or about 3% of the inner core temperatures, which are about 5000°C. Pressure, however, is thought to increase about 11%, from about 329 GPa at the ICB to 364 GPa at earth's center (Fig. 3.8-5). The density inferred from the seismological data is consistent with that for solid iron expected from experiments and modeling.

This situation requires that the inner core geotherm be at temperatures below the melting temperature curve (solidus), whereas the outer core geotherm must be above the solidus.

[10] Iron meteorites — and thus presumably the solid inner core — are like steel, recalling legends in which swords forged from meteorites are very strong and have magical powers.

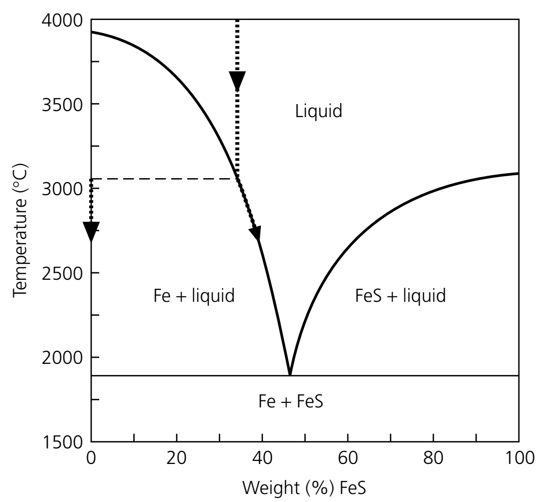


**Fig. 3.8-14** Melting relations for the Fe–FeS system at the pressure of the core–mantle boundary (1.4 Mbar). When a cooling liquid with 33% FeS reaches the phase boundary, solid Fe freezes out, enriching the liquid in FeS. In this analogy, the inner core is freezing out from, and thus chemically different from, the outer core. (Data from Usselman, 1975.)

Two suggestions have been offered for this effect. If the inner and outer cores were chemically identical (Fig. 3.8-13, *left*), the solidus should rise smoothly with depth. The geotherm would be shallower than the solidus, so that they intersect at the ICB, but steeper than the adiabatic gradient required for convection in the outer core. However, some theoretical calculations suggest that the superadiabatic temperature gradient in the core required for convection would be steeper than the solidus. If so, the solid inner and liquid outer cores can be explained by assuming that the inner core is chemically different from the outer core, and thus has a different melting curve (Fig. 3.8-13, *right*). Thus, only in the inner core does the geotherm lie below the solidus and result in a solid phase.

Figure 3.8-14 illustrates this idea, assuming that the light element in the core is sulfur. In this phase diagram for the Fe–FeS system extrapolated to core conditions, sulphur significantly lowers the melting temperature of iron. Cooling a liquid iron mixture with 12% sulphur, corresponding to 33% FeS, causes solid Fe to freeze out, leaving the liquid richer in FeS.[11] In this analogy, the outer core corresponds to the FeS-rich liquid, and the inner core to the denser Fe solid. The nickel would also preferentially enter the solid phase. Such a model predicts an inner core of approximately 80% Fe and 20% Ni, and an outer core with 86% Fe, 12% S, and 2% Ni. The inner core's freezing is thought to be crucial to the convection in the outer core, because the sinking iron releases gravitational potential energy. It has been estimated that the outer core's convection is driven in approximately equal fractions by this process, the latent heat of the crystallizing inner core, and the loss of primordial heat. An additional contribution might come from radiogenic heat production from potassium or uranium, if either are present.

Such models suggest that the boundary between the inner and outer cores is both a phase boundary and a compositional boundary, like the CMB. The boundary may be quite complex. Some evidence suggests that the attenuation of *PKP-DF* waves is greatest in the outer few hundred km of the inner core, implying that this zone may be somewhat mushy. It has also been suggested that iron crystallizes at the ICB at some latitudes, and dissolves back into the outer core at other latitudes, constrained by magnetic forces. This effect may cause preferential alignment of iron crystals, and thus inner core anisotropy (Section 3.6.6). Seismological studies and experimental and theoretical studies of materials at high pressures and temperatures are being used to investigate these issues.

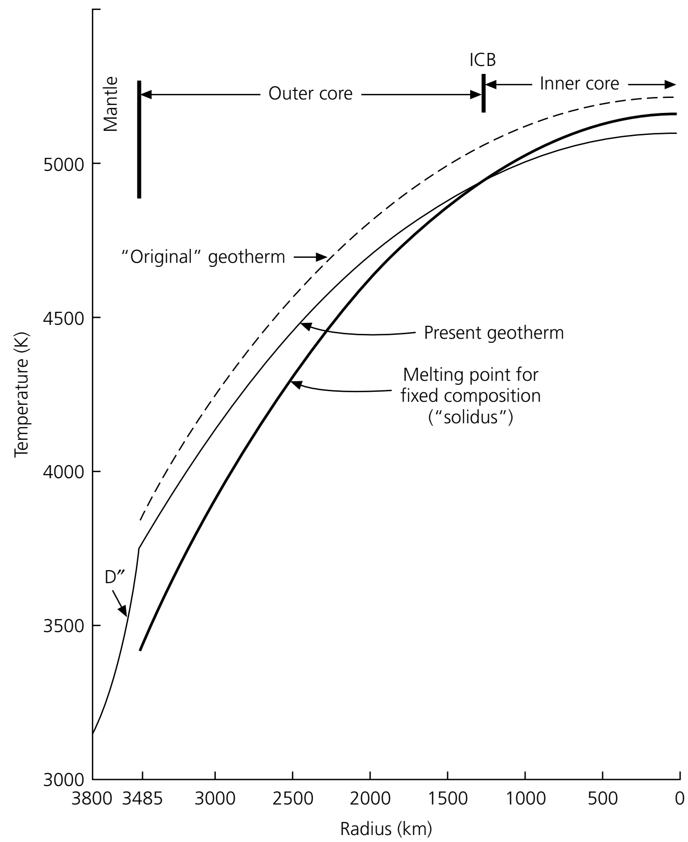


**Fig. 3.8-15** Evolution of the core geotherm, assuming that the solidus is continuous between the inner and outer cores. Early in the earth's history the core geotherm (dashed line) was everywhere greater than the solidus, making the whole core molten. As the core cooled, the geotherm lowered, causing the growth of a frozen inner core. The ICB is the current intersection between the geotherm and the solidus. (After Stacey, 1992. From *Physics of the Earth*, 3rd edn, copyright © 1992 by John Wiley & Sons, Inc. (New York). Reprinted by permission.)

### *3.8.6 Seismology and planetary evolution*

We have seen in this section that seismology gives a snapshot of the present stage of the earth's thermal and chemical evolution. Seismology shows the present thickness of the lithosphere, which may have increased with time, and provides much of our information about plate tectonic processes and mantle convection. Seismology similarly provides most of what we know about the core, including the present sizes of the inner and outer cores that reflect the progressive freezing of the solid inner core from the liquid outer core. Hence, as shown in Fig. 3.8-15, the core has been cooling with time, causing the inner core to grow.

What we know about the earth and our more limited knowledge of the moon and other planets suggest that although there are differences among the inner planets that reflect their initial compositions, there are also similarities in their evolution. As shown in Fig. 3.8-16, planets may follow a similar life cycle, with phases including their formation, early convection and core formation, plate tectonics, terminal volcanism, and quiescence. This evolution is driven by the available energy sources and reflects the planets' cooling with time. Thus, even though the planets formed at about the same time, they are at different stages in their life cycles.[12] The earth is in its middle age, characterized by active plate tectonics.

Hence the approaches used to study the earth's interior can be applied to other planets. A five-station seismological network deployed on the moon by the Apollo missions found a very low level of seismicity, of which most reflected meteoroid impacts or small moonquakes generated by tidal forces. Travel time studies yielded the velocity profile shown in Fig. 3.8-17, which has considerable uncertainty owing to the small number of seismometers and the difficulty of identifying arrivals due to scattering (Fig. 3.7-10). Various interpretations have been made of these results. Although it is tempting to correlate the low-velocity zone with an asthenosphere, thermal models predict that this region would be too cold. As a result, the zonation of the mantle is thought to represent compositional differences.

[11] This effect in which the composition of the liquid and the solid differ is called *fractional crystallization* and has many geological applications, including formation of rocks from a cooling magma. It can be illustrated with partially frozen apple juice, where the liquid tastes sweeter because it is enriched in sugar relative to the solid fraction.

[12] Consider a human and dog born on the same date.

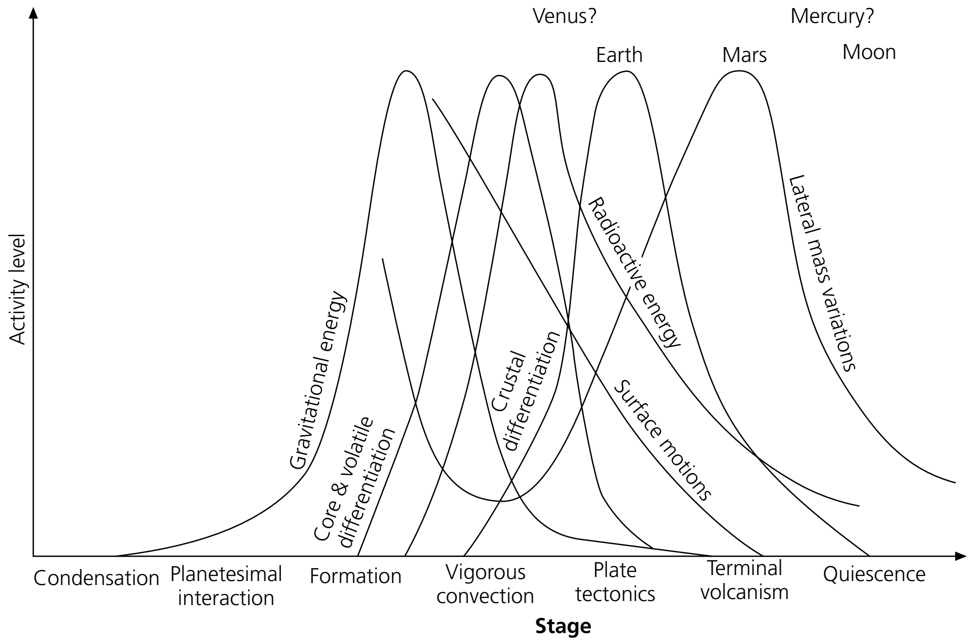


**Fig. 3.8-16** A model for the evolution of the terrestrial planets, showing the energy sources at each stage, presented in text and verse by Kaula (1975). (© 1975 by Academic Press, reproduced by permission of the publisher.)

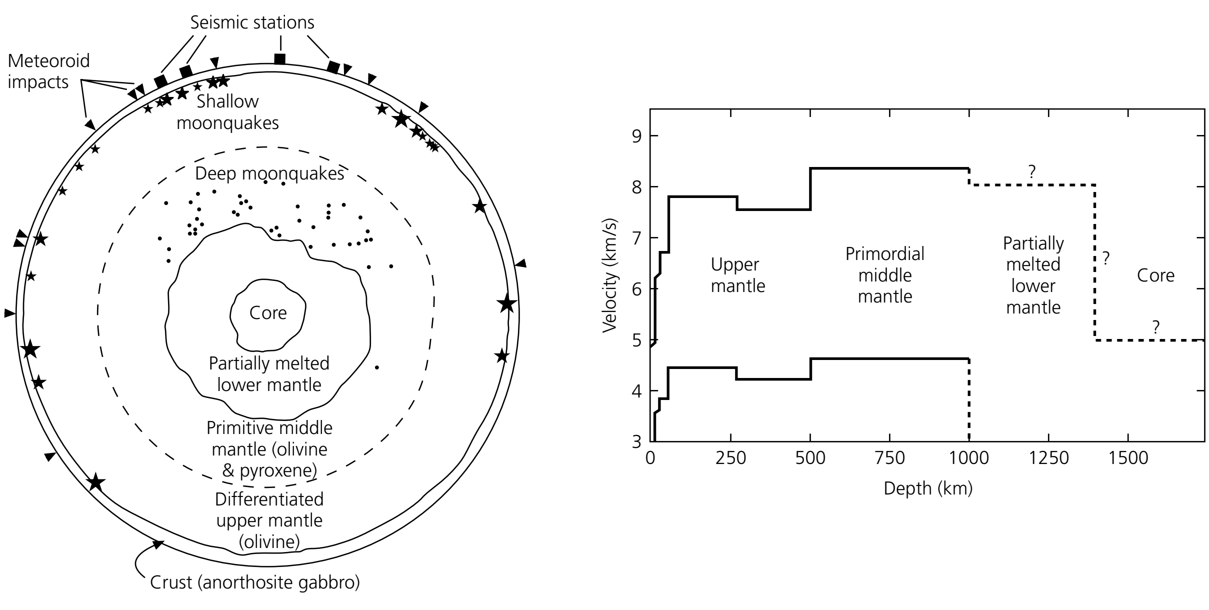


**Fig. 3.8-17** Velocity model from lunar seismic data (*left*) and a possible compositional interpretation (*right*). Squares show seismometer locations, arrowheads show major meteoroid impacts, and large and small dots denote shallow and deep moonquakes. (After Nakamura, 1983 (*J. Geophys. Res.*, *88*, 677–86, copyright by the American Geophysical Union) and Hubbard, 1984.)

There is a suggestion of decreased velocity below 1000 km, which thermal models suggest could be consistent with an asthenosphere. Seismological efforts to detect a core are inconclusive, and the moment of inertia ratio of 0.39 allows for at most a small core.

Hence it appears that the moon now has a thick lithosphere and is tectonically inactive. It thus seems to have lost much of its heat, presumably because of its small size, which favors rapid heat loss. In general, we would expect the heat available from the gravitational energy of accretion and radioactivity to increase as the planet's volume, whereas the rate of heat loss through the surface should depend on its surface area. Hence the remaining heat should vary as

$$\text{remaining heat} = \frac{\text{available}}{\text{loss}} = \frac{(4/3)\pi r^3}{4\pi r^2} = \frac{r}{3}, \qquad (19)$$

so larger planets would retain more heat and be more active.

From such arguments, we might expect Mercury and Mars, which are larger than the moon but smaller than the earth, to have also reached their old age with little further active tectonics. Mercury may still have a small liquid core, which contributes to the observed magnetic field, due to tidal forces from the sun. Venus, which is comparable in size to the earth, might still be active but with episodic, rather than continuous, plate tectonics. Seismology can contribute little to the active discussion of these topics until seismometers are deployed on these planets. Although only one seismometer has been operated on Mars and yielded inconclusive results,[13] seismometers are planned for future missions.

## Further reading

Refraction seismology and its use in crustal studies are covered in many general geophysics texts, such as Fowler (1990) and Reynolds (1997). More detailed treatments can be found in exploration textbooks like Dobrin and Savit (1988), Sheriff and Geldart (1982), Telford *et al.* (1976), and Kearey and Brooks (1984). Additional information can be obtained in review papers such as Braile and Smith (1975), Kennett (1977), or Spudich and Orcutt (1980). A summary of crustal structure results and interpretations for the continental USA can be found in Pakiser and Mooney (1989). Meissner (1986) presents an integrated treatment of observations and models for the continental crust. Reviews on the nature of the Mohorovičić discontinuity are given by Jarchow and Thompson (1989), Braile and Chiang (1986), and Fountain and Christensen (1989). Gibson and Levander (1988) discuss posible artifacts in lower crustal reflection data.

The extensive literature on reflection seismology includes the introductory exploration texts listed above and advanced treatments, including Claerbout (1976, 1985), Robinson and Treitel (1980), Waters (1981), Sheriff and Geldart (1982), Robinson (1983), and Yilmaz (1987). The subject is closely allied to that of geophysical signal processing, discussed in texts including Kanasewich (1981) and Hatton *et al.* (1986).

Applications of seismology to earth structure are discussed in texts and the research literature. Introductory texts such as Bolt (1982), Bott (1982), Gubbins (1990), Doyle (1995), Lay and Wallace (1995), Lowrie (1997), Shearer (1999), and Udias (1999), have good overviews. Simon (1981) is a manual for seismogram interpretation, showing examples of records for earthquakes at various distances and depths. The classic texts by Gutenberg (1959) and Jeffreys (1976) are excellent starting points for further treatment of the data and methods. Bullen and Bolt (1985) has a detailed discussion of ray theory for the spherical earth. Aki and Richards (1980), Ben-Menahem and Singh (1981), and Kennett (1983) treat both ray theory and more advanced methods. The normal mode simulation of body wave propagation shown in Fig. 3.5-19 is available at *http://epsc.wustl.edu/seismology/michael/movie.html*.

Karato and Spetzler (1990) review the physical mechanisms causing anelasticity. Information about anisotropy can be found in Babuska and Cara (1991) and Silver (1996). For discussion of scattering and attenuation, see Kanamori and Anderson (1977), Brennan and Smylie (1981), Jackson (1993), Mitchell (1995), Sato and Fehler (1998), and Romanowicz (1998). Garnero (2000) summarizes results for the lateral heterogeneity of the lowermost mantle.

We alluded only briefly to the nonseismological geophysical data and to chemical results applicable to study of the earth's interior. In addition to journal articles, useful texts are those by Wyllie (1971), Bullen (1975), Ringwood (1975), Wood and Fraser (1977), Brown and Mussett (1993), Bott (1982), Melchior (1986), Jacobs (1987), Lambeck (1988), Anderson (1989), Stacey (1992), and Poirier (2000). Useful reviews can be found in McElhinny (1979), Ahrens (1995a, b, c), Boschi *et al.* (1996), Boehler (1996), Crossley (1997), Gurnis *et al.* (1998), and Davies (1999).

[13] Due to operational constraints, the seismometer was mounted on the lander portion of the spacecraft, rather than in direct contact with Mars. It is rumored that consideration was given to saving weight on the lander by moving the seismometer to the orbiter.

## Problems

1. Use the data from the refraction experiment in Fig. 3.2-5 to find the crust and mantle velocities and the crustal thickness. Remember that this is a reduced travel time plot.
2. For a case of two layers overlying a halfspace, derive an expression for the thickness of the second (deeper) layer in terms of the second crossover distance.
3. Analyze the data from the marine refraction experiment (Lewis, 1978) shown in Fig. P3.1, assuming for simplicity that the structure consists of a water layer, a crustal layer, and a mantle halfspace.
   (a) Assuming that the first arrivals are described by two line segments, for head waves at the top of the crust and mantle, find the corresponding velocities.
   (b) Although the direct wave traveling in the water layer is not shown, the $P$ velocity for water is 1.5 km/s. Use the time intercept for the crustal head wave to find the water depth.
   (c) Use the time intercept for the $P_n$ wave to find the crustal thickness.
4. To show that the head wave is predicted by Fermat's principle, consider a layer of thickness $h$ with velocity $v_0$, overlying a halfspace with a higher velocity, $v_1$.
   (a) Derive the travel time to distance $x$ for a wave that is incident on the boundary at a distance $y$ from the source, travels for some distance just below the boundary, and then returns to the surface at the same incidence angle at which it went down.
   (b) Find the $y$ value giving an extremal travel time, and show that it corresponds to the critical angle of incidence.
   (c) Determine if this travel time is a minimum or a maximum.
5. Use the data for the reversed profile shown in Fig. P3.2 to find the crust and mantle velocities, Moho dip, and crustal thickness.
6. (a) Derive the travel time for the head wave on the up-dip path of a reversed profile with a dipping layer (Eqn 3.2.17).
   (b) Show that the equations for the travel time of the head wave for a dipping layer (Eqns 3.2.16 and 3.2.17) reduce to the flat layer result in the case of zero dip.
7. Derive the Dix equation for interval velocity (Eqn 3.3.19) from the formula for rms velocity.
8. Consider two pairs of seismograms. One pair have the same midpoint, but the offset for one record is the negative of the first. The other pair have the same source point, but the offset for one record